

People analytics project

StaySignal

Most companies find out an employee is a flight risk the day they resign — by then it's too late to act. This project set out to catch that signal early enough to actually do something about it.



employees leaving without warning, and no system flagging who's at risk

The approach

From raw HR data to a live risk score

1

BigQuery

Historical + pilot employee data structured into queryable tables

2

Colab + Random Forest

Trained on past churn patterns, scored the new pilot group

3

BigQuery

Predictions + feature importances persisted back as live tables

4

Looker Studio

Interactive dashboard turning scores into a readable story

Google BigQuery

Google Colab · Python

PyCaret

Random Forest

Looker Studio

What the model found

Clear signals, not a vague guess

2%

of the pilot group
flagged as likely to
leave

26%

importance:
average monthly
hours

22%

importance:
satisfaction level

Risk, quantified

2% of the pilot group was flagged as likely to leave — a small, specific list HR can act on, not a vague company-wide guess.

Drivers, not guesses

Feature importance from the model itself shows average monthly hours and satisfaction level as the strongest predictors of churn.

Where it's concentrated

Risk broken down by department, so retention efforts can target the teams that actually need it.

Why it matters

A model that ends in a decision, not a notebook

Business impact

Turns raw HR data into a short, specific list of at-risk employees and the reasons behind the risk — ready for a retention conversation.

Skills demonstrated

Cloud data warehousing, Python-based ML workflows, model interpretation, and BI storytelling — end to end, one pipeline.

Open to talking about proactive, data-driven people analytics.